PROVENANCE • AUTOMATED PACKING SURVEILLANCE

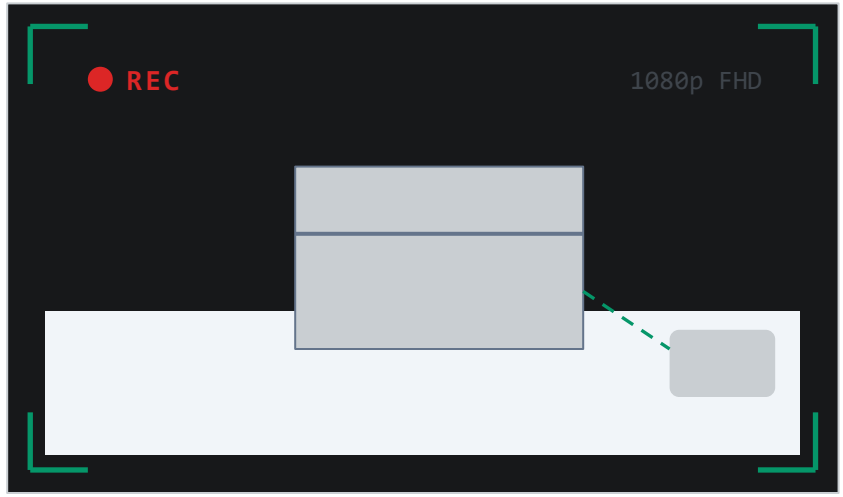
Instant Video Proof for Every Outbound Parcel with Zero Worker Delay

Records automatically from existing handheld barcode scans with zero operator friction.

1080p FHD
Adaptive Vision

0 Clicks
Zero Worker Friction

< 2s
Instant Evidence



CAMERA	1080p Full HD Industrial Sensor
LIGHTING	Dynamic anti-glare optimization
PRE-ROLL	5-second continuous buffer
TRIGGER	Standard barcode scan
STORAGE	Secure local station database

The High-Cost Reality

Unverified Packing Forces Merchants to Forfeit 82% of Claims

Generic ceiling cameras offer zero parcel contents proof, forcing merchants to absorb 100% of dispute losses.

Rs 3,500 to Rs 15,000

Average loss per contested order

82%

Unrecorded claims forfeited by merchants

Stage 01 • Day 0

Blind Dispatch

Operator scans shipping label and packs order. 20-foot ceiling camera is blocked by worker shoulders.

Zero visual record of package contents at seal time.

Stage 02 • Day 5 to 10

Customer Dispute

Buyer claims an empty box or missing item. Marketplace or payment gateway freezes funds automatically.

Merchant carries 100% burden of proof.

Stage 03 • Day 12 to 18

The CCTV Scrub

Staff spends 28 minutes searching generic ceiling footage. Tape glare and blur make contents unidentifiable.

Courier rejects blurry wide-angle clip.

Stage 04 • Day 21+

Claim Forfeited

Courier dispute window closes. Merchant absorbs replacement merchandise, reverse logistics, and penalty charges.

100% of order value permanently written off.

Operational Comparison

Traditional CCTV Takes 28 Mins; Provenance Proves in Under 2 Seconds

Replacing blind ceiling guesswork with instant, package-indexed video proof for every parcel.

Before: Traditional CCTV

Search Time
28 min manual scrub across multiple cameras

Search Accuracy
Guesswork based on estimated dispatch time

Packing Coverage
Misses pre-scan packing steps before label scan

Video Clarity
Blurry wide-angle ceiling view with tape glare

Dispute Win Rate
18% average claim recovery rate

After: Provenance Station Proof

Search Time
Sub-2s instant query by barcode number

Search Accuracy
Exact clip automatically tagged to tracking ID

Packing Coverage
5-second automatic buffer captures item placement

Video Clarity
1080p Full HD bench view with anti-glare tuning

Dispute Win Rate
82% **verified dispute win rate**

Verified packaging video retrieved in seconds reverses false shortage claims with immediate courier sign-off.

Automated Bench Verification

Three Automated Subsystems Capture Proof with Zero Packing Delay

Built specifically for high-speed warehouse packing lines without changing worker habits.

0 Clicks

Zero Worker Friction

Subsystem 01

Instant 0-Click Barcode Trigger

Seamlessly intercepts barcode scans from existing USB or wireless handheld scanners. Triggers recording with zero mouse clicks and zero operator slowdown.

5 Sec

Continuous Buffer

Subsystem 02

5-Second Continuous Visual Buffer

Maintains a rolling visual buffer so the clip captures the moments before the scan, preserving complete footage of item placement and box sealing.

1080p

Adaptive Vision

Subsystem 03

1080p Industrial Camera with Lighting Tuning

High-definition bench camera automatically adjusts exposure and contrast to eliminate tape glare and maintain sharp package clarity in any warehouse lighting.

Three-Step Packing Cycle

Operators Follow a Simple 3-Step Routine Without Touching a Mouse

Every action triggers through physical barcodes, keeping packing flow at full operational speed.

Step 01

Scan Barcode

Operator scans shipping label with handheld scanner. System triggers automatically with 5s pre-roll buffer.

0.2s Duration

Step 02

Pack & Seal Parcel

Bench camera records in 1080p Full HD while worker packs. Visual on-screen indicator confirms active recording.

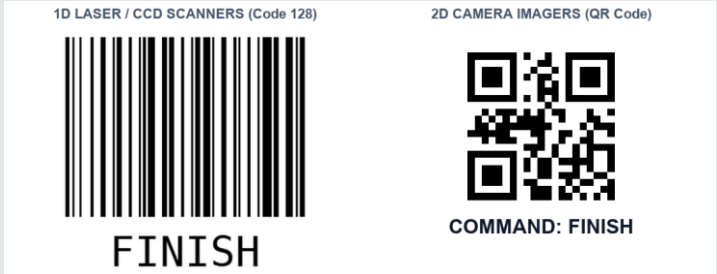
Natural Duration

Step 03

Scan Finish Placard

Operator scans physical desk finish placard. Recording finalizes, saves, and indexes instantly by tracking barcode.

0.1s Duration



Bench Finish Placard • Durable Desk Mount

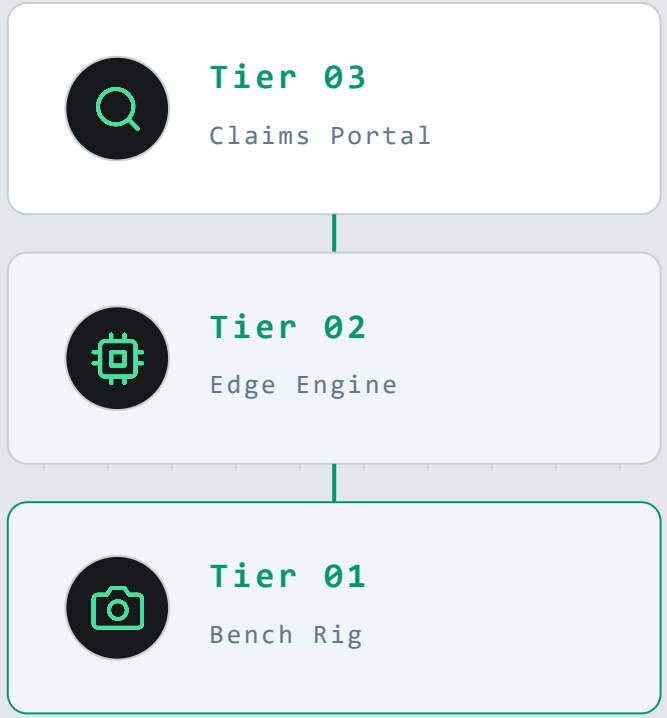
A durable barcode placard sits on the packing table within easy reach. One quick scan finalizes the recording and readies the station for the next order.

No mouse. No keyboard. No window switching.

Local-First Architecture

Local-First Edge Architecture Runs 24/7 Without Cloud or Wi-Fi Dependencies

Video buffers and indices remain 100% operational on local hardware, immune to warehouse network drops.



Dispute & Claims Portal

Instant barcode search by AWB or order ID. Generates a compact MP4 video proof clip ready for one-click submission to courier and marketplace dispute portals.

Sub-2s

Barcode Search

Station Edge Engine

Local-first background service that records, buffers, and indexes video locally on the bench PC. Runs 24/7 with zero dependency on warehouse cloud or network uptime.

100%

Offline Resilient

Hardware Bench Rig

Industrial 1080p camera with dynamic lighting tuning, existing handheld barcode scanner, and clear audio-visual packing confirmation.

1080p

Bench Optimized

Industrial Duty Rig

Heavy-Duty Hardware Built for Continuous Warehouse Vibration and Noise

Rigid industrial-grade components ensure uninterrupted 24/7 performance on active packing tables.

01 • Bench Camera

1080p

1080p Full HD industrial sensor
Adaptive lighting tuning
Anti-glare tape optimization
Wide bench angle coverage
Rigid industrial mount arm

02 • Barcode Scanner

IP65

IP65 industrial casing
USB and wireless support
1D and 2D barcode ready
Fast reads on torn labels
Zero keyboard typing

03 • Station PC

16GB

Dust-sealed fanless chassis
Quad-core edge processor
16GB RAM visual buffer
Fast local solid state drive
Auto 90-day clip storage

04 • Bench Feedback

85dB

Live on-screen indicator
Audible start/stop tones
Heard over warehouse noise
Desk-mounted finish card
Zero worker ambiguity

Financial Business Case

Station Investment Pays Back in Under 90 Days by Reversing False Claims

Verified financial return across high-volume distribution centers without hardcoded assumptions.

Capital Payback Timeline

< 90 Days

Station investment typically recovered through reversed false claims within the first quarter of deployment.

82%

Dispute Win Rate

Recovers 82% of contested claims with indisputable video proof.

15x

Faster Resolution

Resolves claims in seconds instead of 28 minutes of manual scrubbing.

Operational Value Driver	Measured Impact
False Shortage Claims	80%+ contested value recovered
CCTV Investigation Labor	90%+ footage hunting time saved
Courier Dispute Liability	Immediate reversal with video proof
Net Operational Impact	**Positive cash-flow from Month 1**
Capital Payback Horizon	**Full payback in 60 to 90 days**

Turnkey Deployment

Full-Facility Deployment in Four Weeks with Zero Packing Line Downtime

Turnkey implementation designed to fit seamlessly into active packing bench operations.

Week 01 — Week 02 — Week 03 — Week 04

Bench Audit & Setup

Industrial camera bench mount
Handheld barcode integration
Field-of-view & lighting tuning
Clean station power & cabling

01

Station Pilot Launch

Single packing bench activation
5s visual buffer validation
Worker zero-friction observation
90-day local clip storage check

02

Full Facility Expansion

Multi-station facility rollout
Central network storage backup
Claims team portal onboarding
Dispute workflow dry-run

03

WMS & Courier Integration

WMS barcode metadata linking
One-click courier video export
Executive ROI dashboard live
Full production sign-off

04

Next Steps & Evaluation

Deploy Provenance on a Live Packing Bench to Eliminate Dispute Losses

Schedule a zero-risk 30-day pilot or inspect sample courier evidence exports.

Action 01

15-Minute Bench Demo

Inspect the live 1080p camera view, automated barcode trigger, and instant dispute search on a live packing bench.

Schedule: Virtual or on-site walkthrough

Action 02

30-Day Single-Station Pilot

Deploy Provenance on your highest-volume packaging line with full hardware support and zero IT infrastructure overhaul.

Commitment: Zero-risk pilot evaluation

Action 03

Courier Evidence Review

Inspect sample video exports and tamper-evident packaging clips ready for Delhivery, Bluedart, and marketplace portals.

Deliverable: Complete sample evidence clip

Ready to eliminate packing dispute write-offs?

Book a Live Packing Bench Walkthrough

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